Skills Summary

Two-Step Equations and Checking Solutions

Use this reference while completing your worksheet or when studying.

Skill 1 — Solving Two-Step Equations

Rule — undo in reverse order. An equation of the form $ax + b = c$ was built by multiplying, *then* adding. Take it apart the other way round:

- **Step 1:** add or subtract to remove b from the x-side.
- **Step 2:** divide both sides by the coefficient a.

Whatever you do to one side, do to the other — that is what keeps the equation true.

Example: Solve $4x + 5 = 29$
Step 1. The x was multiplied by 4 and then 5 was added, so peel off the addition first and the multiplication second.
Step 2. Subtract 5 from both sides: $4x = 24$. Now divide both sides by 4.
$\Rightarrow$ $\boldsymbol{x = 6}$

Try it: Solve $-2x + 13 = 3$

check: $\boldsymbol{x = 5}$

(!) Watch out: the sign belongs to the coefficient. In $-2x + 13 = 3$ you divide by -2, not by 2 — and a negative divided by a negative gives a positive answer.

Skill 2 — When the Variable Is Divided

Rule. In $\dfrac{x}{a} + b = c$ the variable is *divided*, so the last move is to *multiply*. Clear the added number first, then multiply both sides by a. Division and multiplication undo each other exactly as addition and subtraction do.

Example: Solve $\dfrac{x}{5} + 2 = 7$
Step 1. The x is divided by 5 and then 2 is added, so remove the added 2 first and undo the division last.
Step 2. Subtract 2 from both sides: $\dfrac{x}{5} = 5$. Multiply both sides by 5.
$\Rightarrow$ $\boldsymbol{x = 25}$

Try it: Solve $\dfrac{n}{3} - 4 = 2$

check: $\boldsymbol{n = 18}$

Skill 3 — Checking a Solution by Substituting

Rule. A solution is a number that makes the two sides *equal*. To check one, put it back into the **original** equation, work out the left side on its own, and compare with the right side. Equal means correct; not equal means go back and find the step that broke the balance.

Example: Is $x = 4$ a solution of $3x - 5 = 7$?
Step 1. Checking means testing the candidate, not re-solving — substitute 4 for x in the left side and see what it gives.
Step 2. $3(4) - 5 = 12 - 5$, and the right side is 7, so the two sides agree.
$\Rightarrow$ left side = **7**

Try it: Is $y = -2$ a solution of $5y + 8 = -2$? Work out the left side.

check: left side = **−2**, so yes